

# Homework 4 Key

Solutions by:  
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1. Let  $p, q \in P_n$  with  $p(x) = \sum_{i=1}^n a_i x^i$  and  $q(x) = \sum_{i=1}^n b_i x^i$ . Let  $c, d \in \mathbb{R}$ .  
Then  $cp(x) + dq(x) = c \sum_{i=1}^n a_i x^i + d \sum_{i=1}^n b_i x^i = \sum_{i=1}^n (ca_i + db_i) x^i \in P_n$ . Then by the Subspace Theorem  $P_n$  is a subspace of  $P$ .

2. a. Let  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 0 & -1 \\ -2 & -3 \end{bmatrix}$ .  $\det(A) = -2$  and  $\det(B) = -2$  so  $A$  and

$B$  are invertible. However  $A+B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  and  $\det(A+B) = 0$  so  $A+B$  is not

invertible so the invertible matrices are not a subspace.

b. Let  $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  and  $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ .  $\det(A) = 0 = \det(B)$  so  $A$  and  $B$

are singular. However  $A+B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$  which is clearly

nonsingular. Thus, the singular matrices do not form a subspace.

3. a. No since there are not enough elements.

b.  $\begin{bmatrix} 1 & 6 & 4 & 2 \\ 2 & 3 & -1 & -5 \\ -1 & 0 & 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$  No. The dimension of the span is 2.

c.  $\begin{bmatrix} 2 & -1 & 0 \\ 2 & -2 & 1 \\ 3 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  Yes.

- d. Yes, the standard basis vectors are in the set.

4. a.  $\begin{bmatrix} 1 & 3 \\ 2 & 2 \\ -1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$  Linearly independent.

b.  $\begin{bmatrix} 4 & 2 & 1 \\ 2 & 6 & -2 \\ 1 & -5 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$  Linearly dependent.

$$\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 2 \\ 6 \\ -5 \end{bmatrix}$$

c. Linearly dependent since 4 vectors in  $\mathbb{R}^3$ .

$$\begin{bmatrix} 1 & 0 & 1 & 3 \\ 1 & 2 & 2 & 6 \\ 0 & 3 & 3 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 6 \\ 6 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

d.  $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  Linearly independent.

5.  $0 = c_1 \vec{w}_1 + c_2 \vec{w}_2 + c_3 \vec{w}_3 = c_1(\vec{v}_1 + \vec{v}_2 + \vec{v}_3) + c_2(\vec{v}_2 + \vec{v}_3) + c_3 \vec{v}_3$   
 $= c_1 \vec{v}_1 + (c_1 + c_2) \vec{v}_2 + (c_1 + c_2 + c_3) \vec{v}_3$

Since  $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$  is linearly independent then

$$c_1 = 0$$

$$c_1 + c_2 = 0 \Rightarrow c_1 = c_2 = c_3 = 0.$$

$$c_1 + c_2 + c_3 = 0$$

Thus,  $\{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$  is linearly independent

6.  $\begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 2 & -3 & 1 & 1 & 0 \\ 1 & 1 & 2 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$$7. a. B = \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$$

The dimension is 3.

$$b. B = \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \right\}$$

The dimension is 6.

$$c. B = \left\{ \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \right\}$$

The dimension is 3.

$$8. \begin{bmatrix} 1 & -1 & 2 & 1 & 0 \\ 2 & 0 & 1 & -1 & 3 \\ 5 & -1 & 3 & 0 & 3 \\ 4 & -2 & 5 & 1 & 3 \\ 1 & 3 & -4 & 5 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -3 & 6 \\ 0 & 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{array}{lcl} x_1 & = & 0 \\ x_2 + 3x_4 - 6x_5 & = & 0 \\ x_3 - x_4 + 3x_5 & = & 0 \end{array} \quad \begin{array}{l} x_1 = 0 \\ x_2 = -3s + 6t \\ x_3 = s - 3t \\ x_4 = s \\ x_5 = t \end{array}$$

$$B = \left\{ \begin{bmatrix} 0 \\ 3 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -6 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$9. \begin{bmatrix} 2 & -3 & -7 & 11 \\ 3 & -1 & -7 & 13 \\ 1 & 2 & 0 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 2 & 3 & 1 \\ -3 & -1 & 2 \\ -7 & -7 & 0 \\ 11 & 13 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{row space of } A: \left\{ \begin{bmatrix} 2 \\ -3 \\ -7 \\ 11 \end{bmatrix}^T, \begin{bmatrix} 3 \\ -1 \\ -7 \\ 13 \end{bmatrix}^T \right\} \quad \text{row space of } A: \left\{ \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}^T, \begin{bmatrix} -3 \\ -1 \\ 2 \end{bmatrix}^T \right\}$$

$$\text{column space of } A: \left\{ \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -3 \\ -1 \\ 2 \end{bmatrix} \right\} \quad \text{column space of } A: \left\{ \begin{bmatrix} 2 \\ -3 \\ -7 \\ 11 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ -7 \\ 13 \end{bmatrix} \right\}$$

row space  $A = \text{col space of } A^T$  and col space  $A = \text{row space } A^T$